

Project Proposal

Classification and Prediction On Fruits Dataset

6CS012

Artificial Intelligence and Machine Learning

Herald College Kathmandu

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1. Research Question

Can a Convolutional Neural Network accurately distinguish between 15 specific fruit varieties using a controlled subset of the Fruits-360 dataset?

2. Title

A Performance Analysis of Deep Learning Architectures for the Classification of 15 Selected Fruit Varieties using CNNs and Transfer Learning

3. Dataset Description

Source

The data for this project is a specialized subset derived from the Fruits-360 dataset, publicly available on Kaggle which is authorized by Mihai Oltean. The original dataset is widely recognized in the machine learning community for its high-quality, standardized images of produce.

Dataset Composition

Number of Classes: 15

Total number of Images: 10,192

Training: 7,635 and Test : 2,557

Classes and their Image count:

SN no	Classes (Kaggle dir name)	Training Data	Test Data
1	Apple (Apple 8)	687	228
2	Banana (Banana Lady Finger 1)	450	152
3	Blueberry (Blueberry 1)	462	154
4	Carambola (Carambola 1)	490	166
5	Dates (Dates 1)	490	166
6	Fig (Fig 1)	702	234
7	Guava (Guava 1)	490	166
8	Kiwi (Kiwi 1)	466	156
9	Mango (Mango 1)	490	166
10	Orange (Orange 1)	479	160
11	Pepper (Pepper 2)	480	158
12	Pineapple (Pineapple 1)	490	166
13	Rambutan (Rambutan 1)	492	164
14	Strawberry (Strawberry 1)	492	164
15	Watermelon (Watermelon 1)	475	157

Image Format: All images are 100*100 pixels in RGB color, provided in JPG format.

Collection And Preparation Method

Since, We sourced the dataset from Kaggle, the following preparation was applied:

Extraction Method:

We manually filtered the larger fruits 360 library to extract only 15 categories.

Labeling Approach:

The dataset uses automated labeling based on directory structures. Each image is stored in a folder named after its fruit type, which serves as the ground truth label for the classification algorithm.

Image Acquisition:

The original authors captured these images by placing fruits on a rotating motor at 3rpm and filming them for 20 seconds under consistent lighting against a white background.

Challenges

Class Imbalance:

Some fruits have more images than other fruits (702 vs 450), requiring careful monitoring during model training to prevent bias.

To address class imbalance in the dataset, data augmentation techniques will be applied to increase the number of samples for each fruit class to 1000. In this approach, photometric transformations, specifically Gaussian noise and blurring will be used to generate realistic variations of existing images.

Gaussian noise was added to simulate sensor noise commonly found in real-world image capture. This was achieved by adding random values drawn from a normal distribution to the image pixels, followed by clipping to maintain valid pixel ranges. Additionally, a blurring operation was applied to mimic slight camera defocus, helping the model generalize better to imperfect image conditions.

Inter-class Similarity:

Some fruits are distinct but not every fruit is distinct. Some fruits have similar color profiles like Blueberry and Fig, similar shapes like Guava and Mango, which challenge models to learn textures and shapes rather than color.

Preprocessing Requirements:

Since the images are from a “360-degree” capture, meaning many images are of the same fruit. We had to ensure the split between training and testing data was handled correctly to avoid data leakage.

4. Model Pipeline

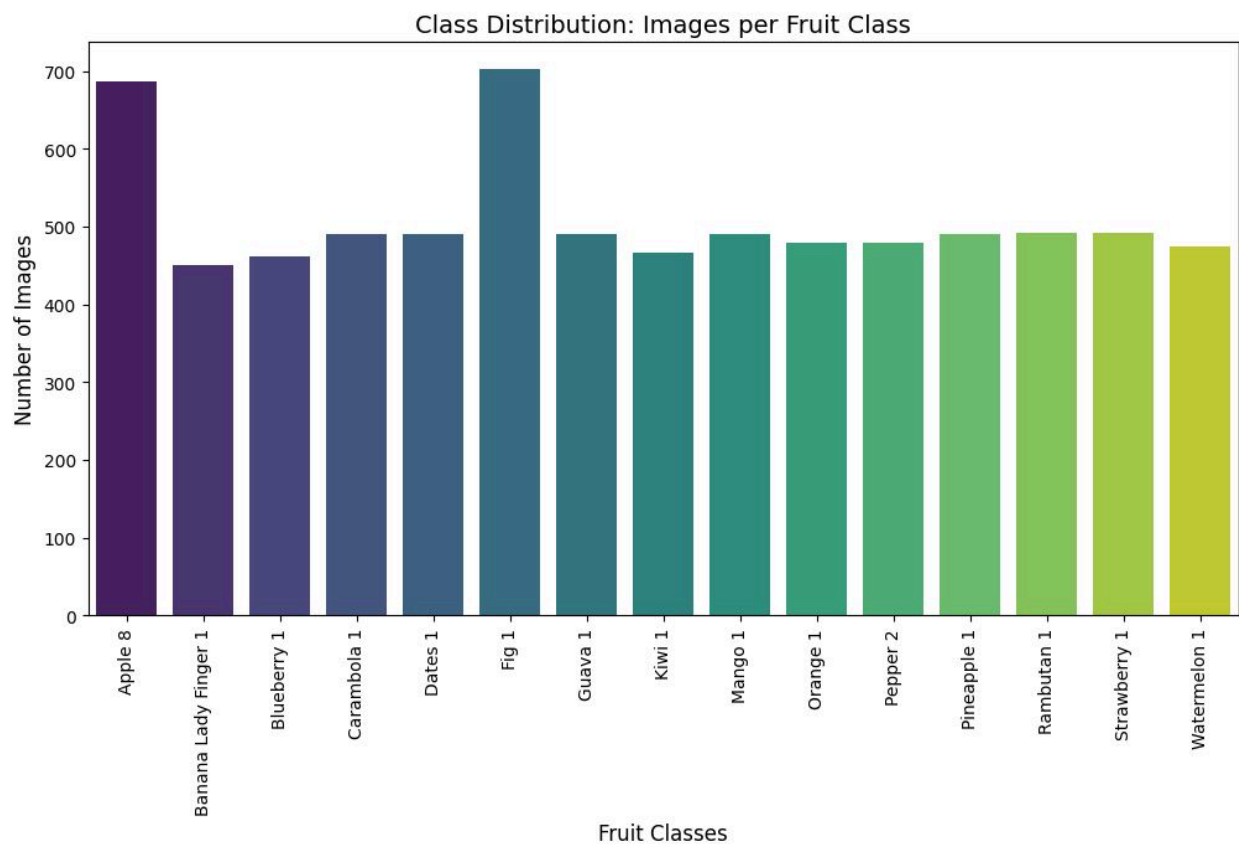
We will be building three models:

- A baseline CNN model
- A Deeper Architecture with Regularization Layer
- Fine-Tuning a Pre-Trained Model (Transfer Learning)

5. Preliminary Data Analysis

Class distribution

Visualizing number of images in each classes



Sample Visualization

Displaying one image from each 15 classes

Sample Images from Selected 15 Fruit Classes

